



1. Create a new node and store the data.
2. Traverse the list to the node after which insertion is required.
3. Set the new node's Next to the next node.
4. Update the previous node's Next to the new node.

Result:

The new node is inserted at the required position.

Advantages of Insertion in Linked List

- No shifting of elements is required
- Memory allocation is dynamic
- Efficient compared to array insertion

Multiple Choice Questions

1. Binary Search works on
 - a) Unsorted array
 - ☒ b) Sorted array
 - c) Linked list
 - d) Stack
2. Sorting is the process of
 - a) Deleting elements
 - ☒ b) Arranging elements in order
 - c) Searching elements
 - d) Copying elements
3. Bubble Sort compares
 - a) Random elements
 - ☒ b) Adjacent elements
 - c) First and last
 - d) Middle elements
4. Big-O notation represents
 - a) Best case
 - b) Average case
 - ☒ c) Worst case
 - d) Constant case
5. A singly linked list node has
 - ☒ a) One pointer
 - b) Two pointers
 - c) No pointer
 - d) Three pointers



6. Time complexity measures
 - a) Memory used
 - ☒ b) Time taken by algorithm
 - c) Code size
 - d) Input size
7. Selection Sort selects
 - a) Largest element first
 - ☒ b) Smallest element and places it correctly
 - c) Middle element
 - d) Random element
8. Towers of Hanoi uses
 - a) Loops
 - ☒ b) Recursion
 - c) Arrays
 - d) Pointers
9. Worst-case time complexity of Bubble Sort is
 - a) $O(n)$
 - b) $O(\log n)$
 - ☒ c) $O(n^2)$
 - d) $O(1)$
10. Hash table is used for
 - a) Sorting
 - b) Searching
 - ☒ c) Fast data access
 - d) Recursion
11. Algorithm efficiency refers to
 - a) Program size
 - ☒ b) Execution time and memory
 - c) Syntax
 - d) Comments
12. Merge Sort follows
 - a) Greedy method
 - ☒ b) Divide and conquer
 - c) Backtracking
 - d) Dynamic programming
13. Linked list stores elements in
 - a) Contiguous memory
 - ☒ b) Non-contiguous memory
 - c) Cache memory
 - d) Stack memory



14. Recursion must have
- a) Loop
 - ☒ b) Base case
 - c) Array
 - d) Pointer
15. Time complexity of Binary Search is
- a) $O(n)$
 - b) $O(n^2)$
 - ☒ c) $O(\log n)$
 - d) $O(1)$
16. Insertion Sort inserts element in
- ☒ a) Sorted part
 - b) Unsorted part
 - c) Middle
 - d) Random
17. Prefix Sum technique is used to
- a) Sort array
 - b) Search elements
 - ☒ c) Find range sum
 - d) Reverse array
18. Data structures are needed to
- a) Increase memory usage
 - ☒ b) Manage data efficiently
 - c) Reduce programs
 - d) Avoid algorithms
19. Time complexity of Merge Sort is
- a) $O(n)$
 - b) $O(n^2)$
 - ☒ c) $O(n \log n)$
 - d) $O(\log n)$
20. Binary Search divides array into
- a) Three parts
 - b) Four parts
 - ☒ c) Two halves
 - d) Multiple blocks
21. A linked list consists of
- a) Only data
 - b) Only pointers
 - ☒ c) Data and pointer
 - d) Array elements



22. Sliding Window technique is used to
- a) Sort array
 - b) Reverse array
 - ☒ c) Find sub-array values
 - d) Delete array
23. Big-O notation is used for
- a) Best case
 - ☒ b) Worst case
 - c) Average case
 - d) Constant case
24. Node insertion in linked list requires
- a) Data only
 - ☒ b) Pointer adjustment
 - c) Array shifting
 - d) Sorting
25. Linear Search works on
- a) Sorted array only
 - ☒ b) Unsorted array
 - c) Tree
 - d) Graph
26. Multidimensional array stores data in
- a) Single row
 - b) Single column
 - ☒ c) Multiple rows and columns
 - d) Linked structure
27. Linked list insertion is faster than array because
- ☒ a) No shifting needed
 - b) Uses cache
 - c) Uses index
 - d) Uses recursion
28. Space complexity measures
- a) Execution time
 - ☒ b) Memory usage
 - c) CPU speed
 - d) I/O time
29. A data structure is used to
- a) Store data only
 - ☒ b) Organize data efficiently
 - c) Delete data
 - d) Print data



30. Recursion means

- a) Function calling another function
- b) Function calling itself
- c) Loop execution
- d) Conditional execution

Key:

1-b, 2-b, 3-b, 4-c, 5-a,
6-b, 7-b, 8-b, 9-c, 10-c,
11-b, 12-b, 13-b, 14-b, 15-c,
16-a, 17-c, 18-b, 19-c, 20-c,
21-c, 22-c, 23-b, 24-b, 25-b,
26-c, 27-a, 28-b, 29-b, 30-b

Fill in the Blanks:

1. Selection Sort uses minimum element selection in each Iteration.
2. In recursion, the stopping condition is called Base Case.
3. Hash table uses Hash function.
4. Data structures improve program Efficiency.
5. Bubble Sort pushes largest element to the End.
6. Time complexity depends on input Size.
7. Recursion uses function Calls.
8. Sorting arranges data in Ascending or descending order.
9. Binary Search follows Divide and conquer technique.
10. Linear Search compares elements one by one.
11. Multidimensional array example is 2D array.
12. Prefix Sum helps in fast Range sum queries.
13. A data structure organizes data for Efficient processing.
14. Quick Sort complexity is $O(n^2)$.
15. Linked list node contains data and Pointer.
16. Algorithm must have Finite steps.
17. Towers of Hanoi problem has Three rods.
18. Insertion Sort builds Sorted array gradually.
19. Linked list overcomes the limitation of Contiguous memory.
20. Searching in linked list is done Sequentially.
21. Worst-case complexity of Bubble Sort is $O(n^2)$.
22. Space complexity refers to Memory usage.
23. Big-O notation represents Worst case.



24. Insertion in linked list is done by changing Pointers.
25. Sliding Window has fixed window size .Constant.
26. Doubly linked list has Two pointers.
27. Time complexity measures Time taken by algorithm.
28. Linear search works on unsorted array.
29. Merge Sort divides array into Two halves.
30. Linked list node contains data and a Pointer.

Key:

1. Pass or iteration
2. Base case
3. Hash
4. Efficiency
5. End
6. Size
7. Calls
8. Ascending
9. Divide
10. One
11. 2D
12. Range sum
13. Efficient processing
14. $O(n^2)$
15. Pointer
16. Finite
17. Three
18. Sorted
19. Contiguous
20. Sequentially
21. $O(n^2)$
22. Memory usage
23. Worst
24. Pointers
25. Constant
26. Two
27. Time
28. Linear
29. Two
30. Pointer



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